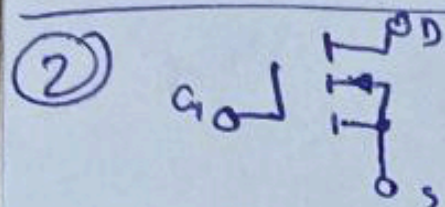
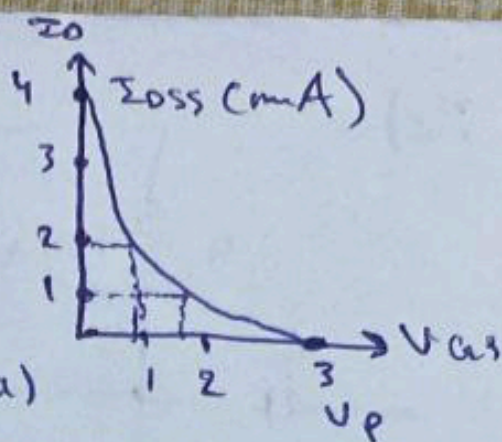


$$① I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$$I_D = 4 \left(1 - \frac{2}{3}\right)^2 \times 10^{-3}$$

$$I_D = 0.444 \times 10^{-3} \rightarrow 0.444 \text{ mA (a)}$$



n-channel E-MOSFET
(c)

$$③ V_{GS(TH)} = 3 \text{ V}, K = 0.061 \text{ mA/V}^2$$

$$I_{D_{on}} = K(V_{GS(on)} - V_{TH})^2$$

$$= 0.061(10 - 3)^2 = 2.989 \approx 3 \text{ (a)}$$

on-state Drain current

$$V_{GS} = 10, V_{DS} = 10$$

Drain Source on Voltage

$$I_D = 2 \text{ mA}, V_{GS} = 10 \text{ V}$$

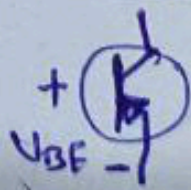
4)

If p-channel JFET transistor, $V_{GS(off)}$ is POSITIVE (a)

β/h_{FE} in the data sheet is β_{dc} (d)

6) Value of I_{DSS} in a JFET is defined $V_{DS} \geq V_P$ and $V_{GS} = 0 \text{ V}$
(c)

7) Value of V_{BE} is (-0.7 V)
(b)

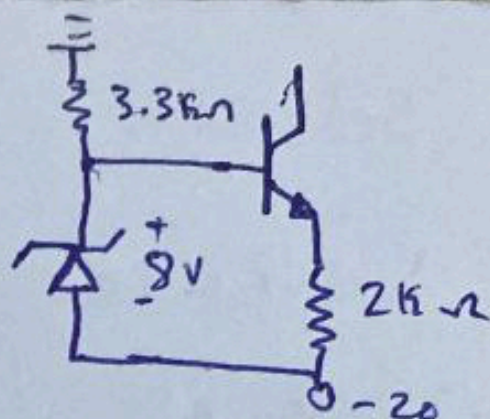


8)

6mA

قيد

(a).



$$a) V_B = \frac{R_2 \cdot V_{cc}}{R_1 + R_2} = \frac{18 \cdot 1.5K}{(4.7 + 1.5)K} = 4.35 (b)$$

10) Correct statement for PNP transistor in active region.

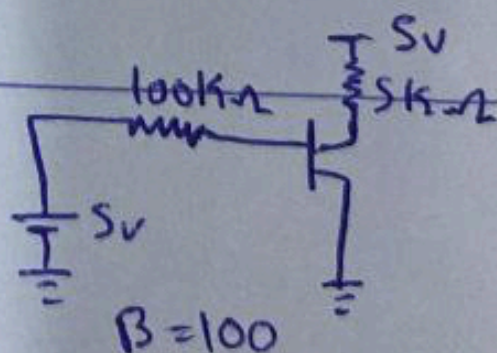
Base-Emitter $\rightarrow$ F.B Base-Collector

$$11) I_{C \text{ sat}} = 1mA. \left\{ \right.$$

$$I_B = 0.043$$

$$I_C = 4.3$$

Saturation (b)



12) Small amount of current flows when base current I_{B20} is (I_{CBO}) (b).

13) If I_C increases, V_{CE} will DECREASE (c)

14) $I_{DSS} = 10 \text{ mA}$ Find β for saturation. (D)

15)

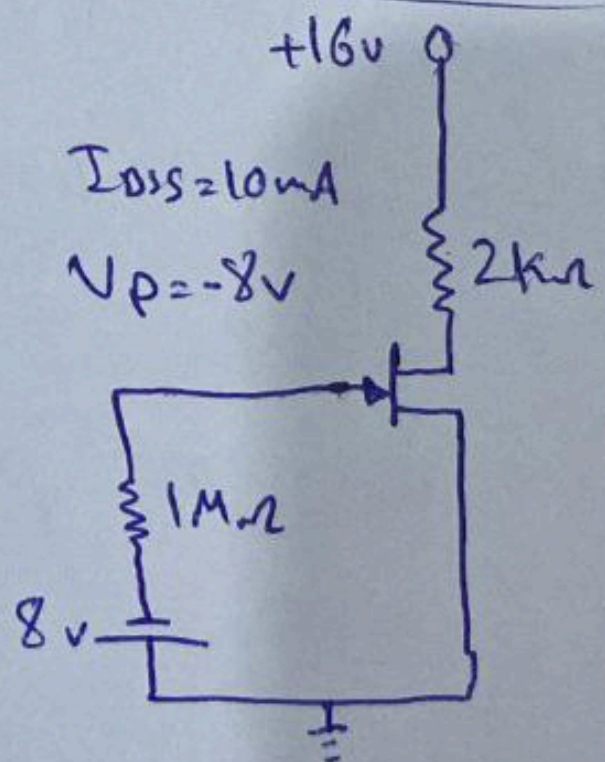
116

$$V_{GS} = -8$$

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$$I_D = 0$$

$$V_{DS} = 0. (a)$$



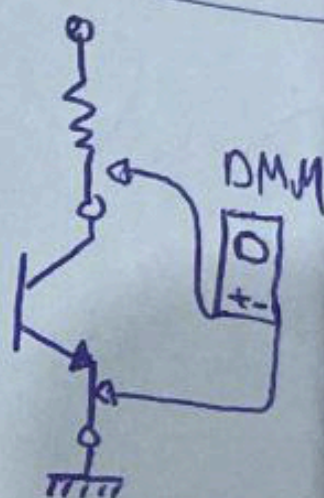
$q \rightarrow 8.7$
 or
 $q \rightarrow 8.7$ (Q1)
 $\checkmark$

16) Two regions that a transistor is used as a switch and has fixed values are: Saturation & Cutoff (C).

17) $V_{CC} = 20V$, $R_C = 5k\Omega$, $I_C = 2mA$

Reading $\rightarrow V_{CE}$

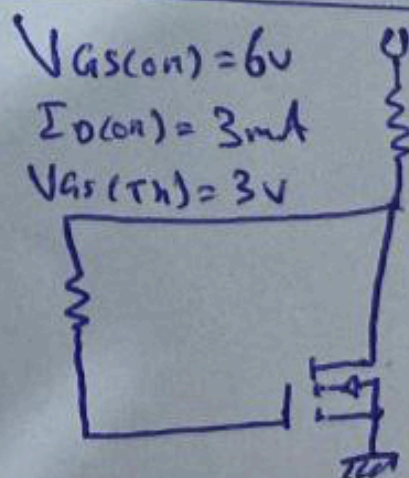
$$V_{CE} = 20 - 5 \times 2 = 10V. (a)$$

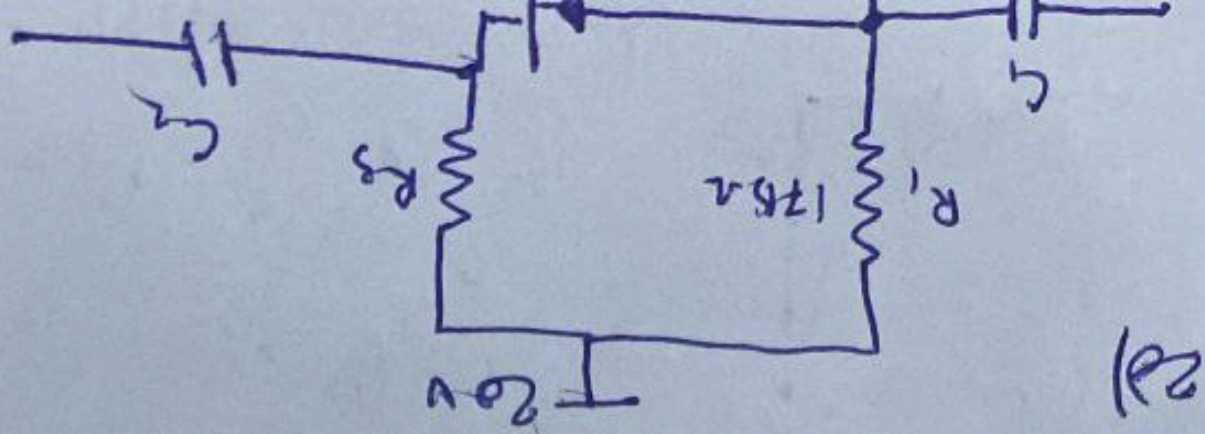


18) A BJT is _____ controlled device
 A JFET is _____ controlled device

(C) Current, Voltage

19) $I_S = \frac{3m}{(6-3)^2} = 0.33mA/V^2$
 (C).





Self Bias n-channel JFET

$$V_{GS} = 0V$$

$$I_D = 4mA$$

$$V_{GS} = -4V$$

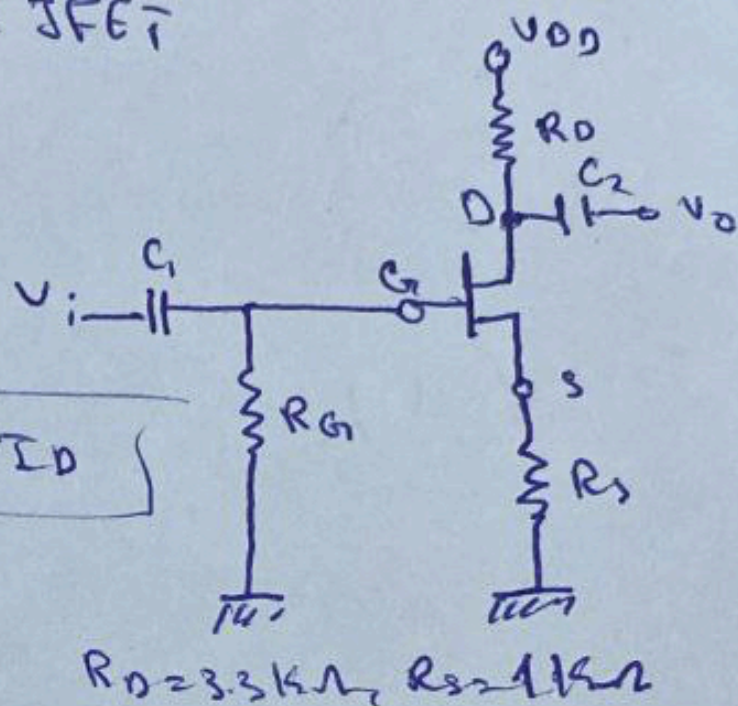
$$V_{GS} = -I_D R_S$$

$$V_D = V_{DD} - I_D R_D$$

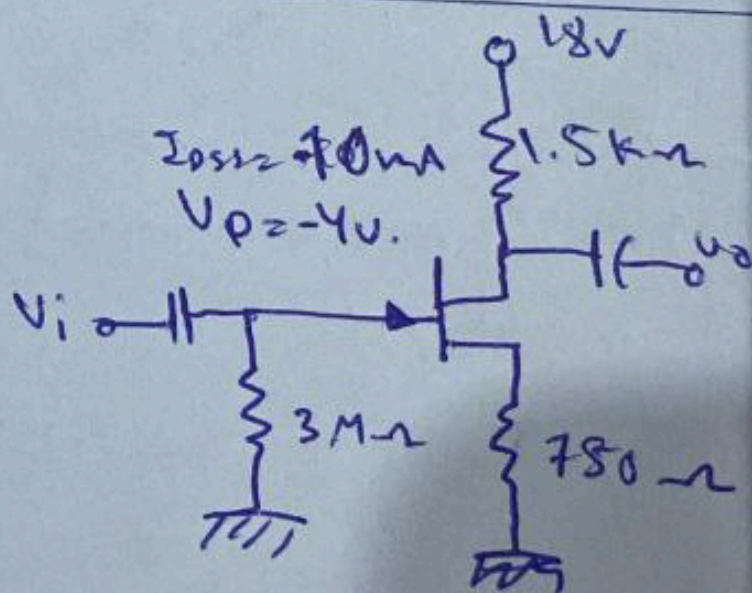
$$= 20 - (4 \times 3.3)$$

$$= 11.42$$

$$V_{DD} = 20, I_{DSS} = 8, V_P = -6, R_G = 1M\Omega$$



$$R_D = 3.3K\Omega, R_S = 1K\Omega$$

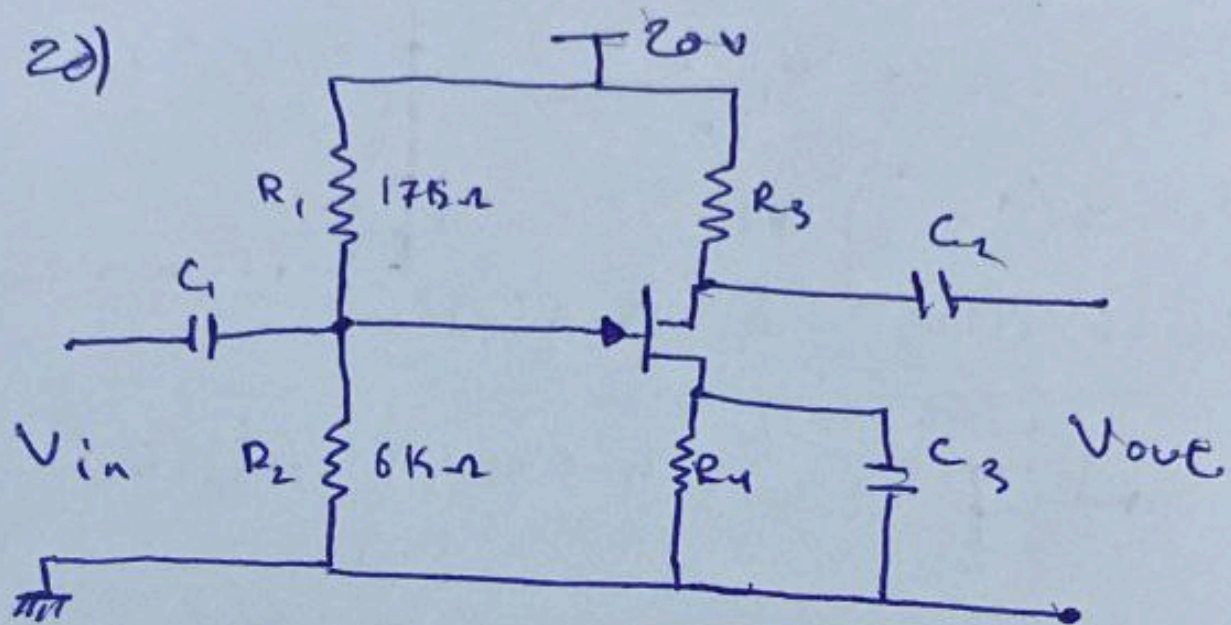
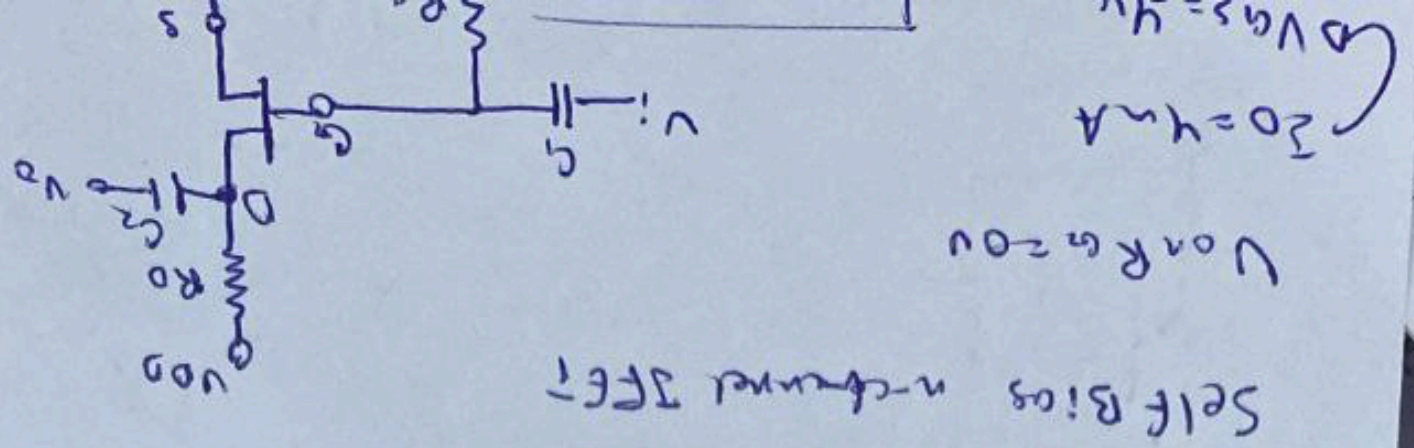


$$I_{DSS} = 10mA$$

$$V_P = -4V$$

$$3M\Omega$$

$$780\Omega$$



$$V_G = V_{R_2}$$

$$V_G = \frac{20 \times 6}{17 + 6} = 5.2 \text{ (a)}$$